

Bending Deflection: δ : the deflection
→ how far
the beam moves
at the point
where the
load is applied

$$\delta = F \cdot L^3 / (48 \cdot E \cdot I)$$

where F is the acting
force, L is the unsupported
distance, E is Young's Modulus
and I is the second moment of
area

$$F_{\text{wt}} = 40 \text{ N for wood, } E = 3 \text{ GPa}$$

(for PETG)

$$\rightarrow 3 \cdot 10^9 \text{ Pa} \Rightarrow$$

$$L = 0.233 \text{ m}$$

for a rectangle: $I = b \cdot h^3 / 12$ where
 b is the dimension perpendicular to
the load and h is the dimension
parallel: $I = 0.008 \cdot 0.083^3 / 12 =$
 $3.8 \cdot 10^{-7} \text{ m}^4$

$$\delta = 1.4 \cdot 10^{-5} \text{ m} = 3.5 \text{ mm}$$

The Euler-Bernoulli Beam eq. tells us that: $E \cdot I y''(x) = M(x) \rightarrow$
where $y(x)$ is the deflection
pos. x along the beam
and $M(x)$ is the bending moment.

$\rightarrow$ we analyse the center of the beam for symmetry, such that each end has a vertical rxn of $F/2$, s.t. for $0 < x < L/2$

$M(x) = (F/2)x - M_0$ where M_0 is to be evaluated; $\int_0^{L/2} M(x) dx =$

$$\frac{1}{2} (F/2)x^2 - M_0 x \Big|_0^{L/2} = F/4 (L/2)^2 - M_0 L/2$$

$$\Rightarrow F \cdot L^2/16 - M_0 L/2 \text{ (rhs)}$$

for the lhs, $E \cdot I y'(x)$, so that:

$$E \cdot I y'(L/2) = F \cdot L^2/16 - M_0 L/2$$

$$\text{if } y'(L/2) = 0, \text{ then: } M_0 = F \cdot L/8$$

if we integrate the eq:

$$E \cdot I y'(x) = (F/4) x^2 - M_0 x \quad \text{we get}$$

$$\hookrightarrow E \cdot I y(x) = F/12 x^3 - M_0/2 x^2 + C$$

where if $x = L/2$ and $C = 0$,
then b/c no bending at $x = 0$

$$y(L/2) = \frac{F/12 (L/2)^3 - M_0/2 \cdot (L/2)^2}{E \cdot I}$$

if $M_0 = F \cdot L/8$, then:

$$F/12 \cdot L^3/8 - F \cdot L/16 \cdot L^2/4$$

$$\frac{F \cdot L^3}{96} - \frac{F \cdot L^3}{64} \Rightarrow \frac{2F \cdot L^3}{192} - \frac{3FL^3}{192}$$

$$E \cdot I \Rightarrow - F \cdot L^3 / (192 \cdot E \cdot I)$$

$\hookrightarrow$ Remember that $y(x)$ is the deflection, and that it deflects in the opp. direction.

$$\boxed{\delta = F \cdot L^3 / (192 \cdot E \cdot I)}$$